Token vs Context in LLM Representations

MATH 598 Final Project

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May 2026

Abstract

Transformer hidden states combine information from the current token and from the preceding context. This project measures how that balance changes across layers by asking the following question: how well can a layer's hidden state be predicted from the token's embedding alone? Using a token-only reconstruction probe, for each held-out token, an affine map is trained to predict a layer's hidden state from the token's input embedding alone. I apply this method to EleutherAI Pythia-70M and Pythia-160M on samples from NeelNanda Pile-10k. In both models, the first layer is strongly recoverable from token identity, while middle layers become much less predictable. The Pythia-160M run shows a clear early drop from $R^2 = 0.76$ at layer 0 to about 0.11 across layers 3–8, suggesting that most token-only linear structure is lost early in the network. The final layer in both models has unusually high cosine similarity despite large mean squared error, which appears to be a scale effect rather than evidence of complete reconstruction. Overall, the results support the hypothesis that context-dependent information becomes dominant after the earliest transformer blocks, while also showing the importance of reporting multiple metrics when probing hidden representations.

1 Introduction

Large language models (LLMs) represent each token using both the token itself and the tokens that came before it. At the input layer, the model starts with an embedding lookup that depends only on the token id. After attention and feed-forward blocks, though, the hidden state at that same token position can include information from earlier words in the sequence. This project studies where that change shows up: how much of a transformer's intermediate representation can be predicted from the current token alone, and how much appears to depend on the prior context?

The central experiment is a token-only reconstruction test. For each observed token position in the dataset, I collect the model's input embedding and selected hidden states. I then fit a simple affine probe from the token embedding to each layer's hidden state. If the probe predicts a layer well, then that layer's representation is largely recoverable from token identity alone under this probe. If the prediction quality is poor, missing information suggests that the layer has additional information that cannot be recovered from the token alone and thus that it has become more context dependent. This does not prove exactly what the model is doing internally, but it gives a useful way to measure how much of the observed representation remains linearly predictable from a context-free input.

This framing gives a quantitative way to compare layers. Early transformer layers may remain close to lexical or local-token information, while later layers are expected to mix in more contextual structure. A layerwise curve of mean squared error, cosine similarity, and R^2 can therefore show whether token-only predictability decays gradually, drops sharply at a specific depth, or remains unexpectedly high throughout the network.

The project leaves room for a feature-level extension using sparse autoencoders. Hidden-state vectors are useful for measuring overall predictability, but they are hard to interpret directly. Sparse autoencoder features may make it possible to ask a more specific version of the same question: which learned features are token-local, and which features require context? The current implementation focuses first on the hidden-state baseline, with the sparse-feature analysis planned as a second stage once the baseline pipeline is stable.

2 Literature Review

This project builds on work in transformer interpretability, probing, and sparse feature decomposition. The common thread across these areas is the attempt to understand what information is present inside a model's intermediate representations, and how that information changes across layers.

The transformer architecture introduced by Vaswani et al. [9] is the starting point for this question. Self-attention allows each token position to combine information from other positions, so the representation at a position is not limited to the token's own embedding. In causal language models, this mixing is constrained to the current and previous tokens, making the distinction between token-local information and prior-context information especially natural. The experiment in this report uses that distinction directly: the probe receives only the current token embedding, while the target hidden state was produced by a model that had access to the earlier context.

Mechanistic interpretability work on transformer circuits motivates looking for structured computations inside transformer layers rather than treating hidden states as opaque vectors. Elhage et al. [5] describe transformers in terms of residual streams, attention heads, and feed-forward components that can be analyzed as interacting computational units. That perspective supports the idea that layerwise representations may contain a mix of simple token-level components and more contextual computations. Olsson et al. [8] study induction heads and in-context learning, showing how specific attention patterns can support behavior that depends on earlier tokens. This motivates the expectation that later representations should often become harder to recover from token identity alone.

The project is also related to work that asks what predictions are available from intermediate model states. The tuned lens of Belrose et al. [2] learns transformations from intermediate residual streams to the model's output distribution, revealing how model predictions develop across depth. My experiment uses a different direction of prediction. Instead of asking what output distribution is already present in a hidden state, I ask how much of the hidden state can be reconstructed from the context-free token embedding. Both approaches use learned linear maps to make layerwise information measurable, but they emphasize different relationships: hidden state to output in the tuned lens, and token embedding to hidden state here.

Linear probing provides the methodological basis for this measurement. Alain and Bengio [1] use linear classifier probes to study information available in intermediate layers, and Hewitt and Manning [7] show that linear probes can recover syntactic structure from word representations. These works support probing as a practical tool for studying representations, but they also point to an important limitation: probe performance measures what is recoverable by the probe, not necessarily what the model itself uses. In this project, high token-only probe performance means that the hidden state is linearly predictable from the token embedding. Low performance is consistent with context dependence, but it could also reflect a nonlinear relationship or limitations of the affine probe. For that reason, results should be interpreted as evidence about linear recoverability rather than a complete causal explanation.

Sparse autoencoder work provides the motivation for the planned feature-level extension. Bricken et al. [4] use dictionary learning to decompose language model activations into sparse features that can be more interpretable than raw activation dimensions. If suitable sparse autoencoders are available for the chosen model, the same token-only predictability test can be applied to feature activations. This would shift the analysis from asking whether an entire hidden-state vector is token-local to asking which individual features are token-local. That extension would help connect quantitative predictability curves to more interpretable claims about the types of information represented by the model.

Together, these lines of work motivate the project's baseline experiment. Prior work shows that transformer layers develop meaningful intermediate states, that linear probes can expose recoverable information in those states, and that sparse methods may decompose activations into interpretable features. This project uses those ideas to isolate a specific question: how much of a representation can be explained by the current token before any context is used?

3 Methodology

The experiments use two pretrained causal language models from the Pythia suite [3]: Pythia-70M and Pythia-160M. Pythia was chosen because it is open, small enough to run locally on CPU for this project, and designed for controlled analysis across model sizes. The text data comes from Pile-10k, a small Hugging Face sample derived from the Pile [6]. The 70M run uses 25,000 token positions from the first 1,000 examples of Pile-10k; the 160M run uses 50,000 token positions from the Pile-10k training split. Both runs use a maximum sequence length of 128 tokens and remove padding before fitting probes.

For each observed token position i, the extraction code stores two things: the context-free input embedding $x_i \in \mathbb{R}^d$ and the contextual hidden state $h_{i,\ell} \in \mathbb{R}^d$ at layer ℓ. The input embedding depends only on the token id, while the hidden state is produced after causal self-attention and feed-forward layers have had access to previous tokens. This makes the supervised learning problem

$$x_i \mapsto h_{i,\ell}$$

a direct test of how much the hidden state can be recovered from the token alone.

For every layer, I split token positions into 80% train and 20% test sets. To check that the results are not just a consequence of a lucky split, I repeat this process 10 times using random seeds 0 through 9. For each split, I then fit one affine probe per layer,

$$\hat{h}_{i,\ell} = W_\ell x_i + b_\ell,$$

by solving the ordinary least-squares problem on the training split. No ridge regularization is used in the main results. The pipeline code is organized under `src/`: extraction collects model activations, probing fits the affine maps, and diagnostics produces the per-token plots.

I evaluate each probe on held-out tokens with three metrics. Mean squared error (MSE) measures absolute reconstruction error. The coefficient of determination (R^2) compares the probe to a constant mean-target predictor, so values near zero mean the probe adds little beyond the average hidden state. Mean cosine similarity measures directional alignment between predicted and true hidden states. For Figures 1 and 3, the shaded region shows one standard deviation across the 10 splits. Reporting all three metrics matters because cosine similarity can look high even when MSE is large if target vectors have very large norms.

4 Results

4.1 Pythia-70M

Figure 1 shows the Pythia-70M layerwise results averaged across 10 train/test splits, with the shaded region showing one standard deviation. The first two layers are the most token-predictable: in the seed-0 split reported in Table 1, R^2 is 0.70 at layer 0 and 0.47 at layer 1, with mean cosine similarities of 0.89 and 0.81. By layer 2, R^2 drops to 0.16 and remains low through the final block. This is the main evidence that the representation becomes much less linearly recoverable from token identity after the earliest transformer layers.

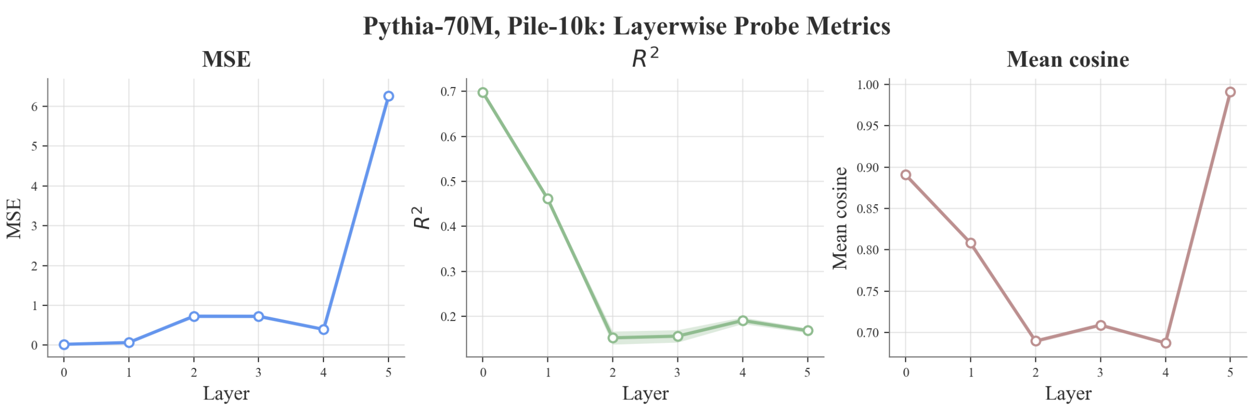

Figure 1: Layerwise token-only probe metrics for Pythia-70M on 25,000 Pile-10k token positions. Lines show the mean across 10 train/test splits, and shaded regions show one standard deviation.

Figure 2 gives a qualitative view of how the probe output behaves on individual examples. These examples are not used as the main evidence, but they help show why the layerwise metrics should be read as an aggregate

Layer	MSE	R^2	Mean cosine
0	0.021	0.699	0.891
1	0.064	0.467	0.808
2	0.676	0.156	0.689
3	0.675	0.160	0.709
4	0.383	0.195	0.688
5	6.065	0.170	0.992

Table 1: Held-out Pythia-70M probe metrics for seed-0 split. The split contains 20,000/5,000 train/test token positions.

pattern rather than as a guarantee that every token behaves similarly. In particular, the example-level view shows that token-only reconstruction can look reasonable for some held-out tokens even after the aggregate R^2 has fallen, which is why the per-layer averages and qualitative examples are useful together.

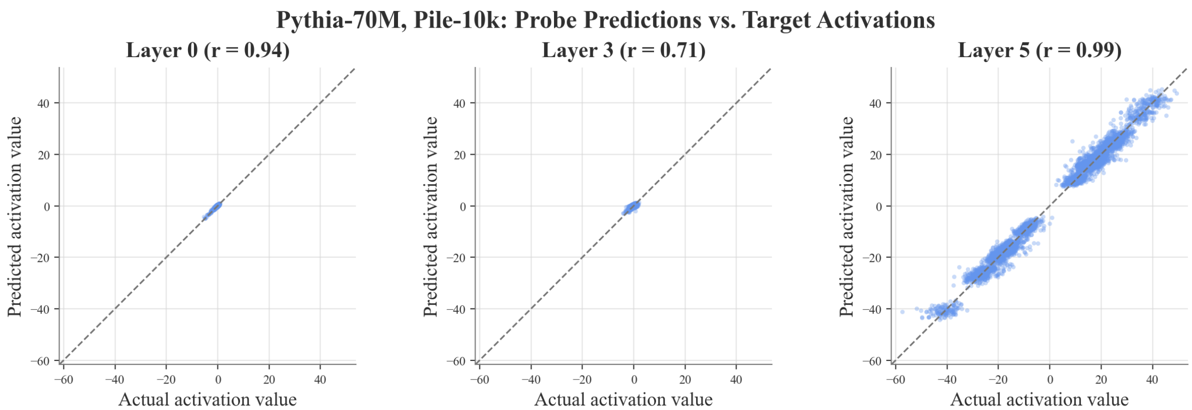

Figure 2: Pythia-70M qualitative prediction examples from the diagnostic pipeline. The examples compare true and reconstructed hidden-state behavior for selected held-out tokens.

The final layer requires special interpretation. Its MSE is much larger than the earlier layers, but cosine similarity rises to 0.99. The diagnostic summary shows that the mean hidden-state norm at layer 5 is about 440, compared with about 7–16 in earlier layers. This suggests that the high cosine is mostly measuring directional alignment in a very large-norm representation, not close coordinate-level reconstruction. Supporting diagnostic figures for this point are included in Appendix B.

4.2 Pythia-160M

The larger model gives a cleaner view because it has 12 layers. As shown in Figure 3, token-only predictability is high at the first layer, declines at layers 1 and 2, and then enters a long middle region where the probe explains very little variance. R^2 falls from 0.76 at layer 0 to 0.52 at layer 1, 0.26 at layer 2, and roughly 0.10–0.13 through layers 3–9. This pattern is consistent with contextual mixing becoming dominant early in the network.

The larger model gives a cleaner view because it has 12 layers. Figure 3 shows the Pythia-160M layerwise results averaged across 10 train/test splits, with the shaded region showing one standard deviation. Token-only predictability is high at the first layer, declines at layers 1 and 2, and then enters a long middle region where the probe explains very little variance. In the seed-0 split reported in Table 2, R^2 falls from 0.76 at layer 0 to 0.52 at layer 1, 0.26 at layer 2, and roughly 0.10–0.13 through layers 3–9. This pattern is consistent with contextual mixing becoming dominant early in the network.

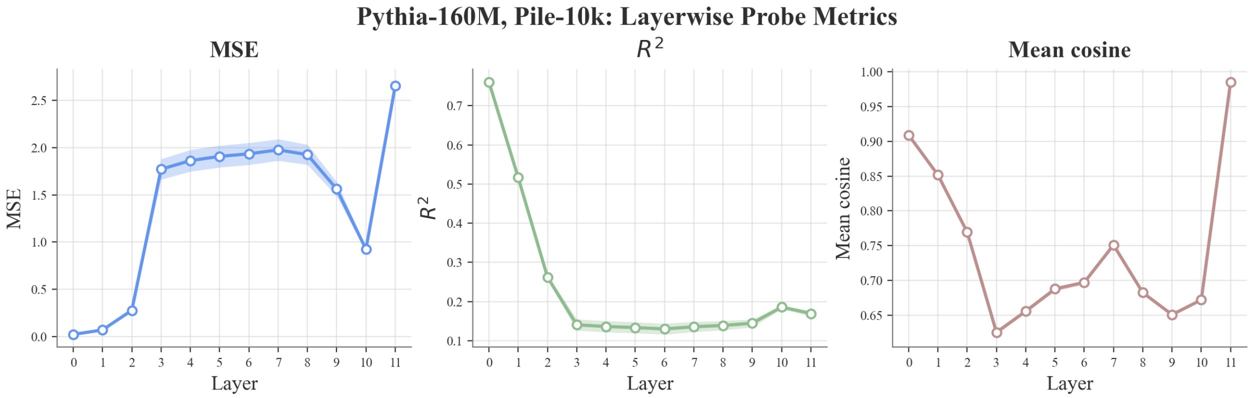

Figure 3: Layerwise token-only probe metrics for Pythia-160M on 50,000 Pile-10k token positions. Lines show the mean across 10 train/test splits, and shaded regions show one standard deviation.

Layer	MSE	R^2	Mean cosine
0	0.021	0.759	0.908
1	0.067	0.519	0.851
2	0.266	0.263	0.769
3	1.715	0.112	0.623
4	1.802	0.108	0.654
5	1.846	0.106	0.686
6	1.872	0.103	0.696
7	1.914	0.110	0.750
8	1.868	0.116	0.681
9	1.519	0.125	0.650
10	0.909	0.177	0.672
11	2.578	0.163	0.986

Table 2: Held-out Pythia-160M probe metrics for the seed-0 split. The split contains 40,000/10,000 train/test token positions.

The same qualitative diagnostic for Pythia-160M appears in Figure 4. Together with the main 160M metric curve in Figure 3, these examples make the result more concrete: early layers remain comparatively token-predictable, while middle-layer behavior varies more across held-out tokens.

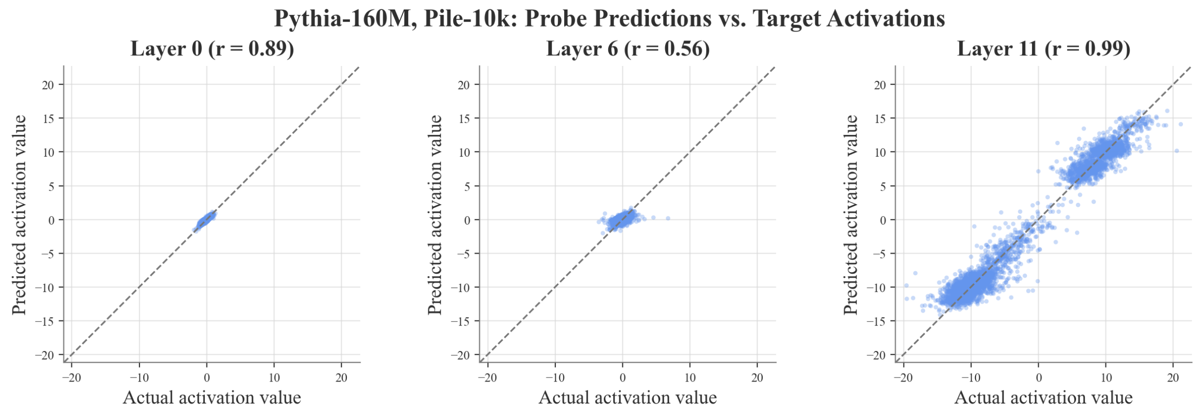

Figure 4: Pythia-160M qualitative prediction examples from the diagnostic pipeline. These examples supplement the aggregate layerwise metrics by showing selected held-out token reconstructions.

The Pythia-160M diagnostics also show that the probe consistently beats a constant mean baseline, but the margin becomes small in the middle layers. For example, layer 6 has probe MSE 1.87 versus mean-baseline MSE 2.09, and layer 8 has probe MSE 1.87 versus baseline MSE 2.11. Thus the token embedding still contains some useful information, but most of the held-out hidden-state variance is not captured by the token-only affine map. The full set of diagnostic plots is included in Appendix B.

4.3 Comparison Across Models

Figure 5 puts the two R^2 curves on a common relative-depth axis. Unlike Figures 1 and 3, this comparison plot is shown as a direct model-to-model comparison rather than as a 10-split average with shaded uncertainty. This makes the shared trend easier to see despite the different number of layers in the two models.

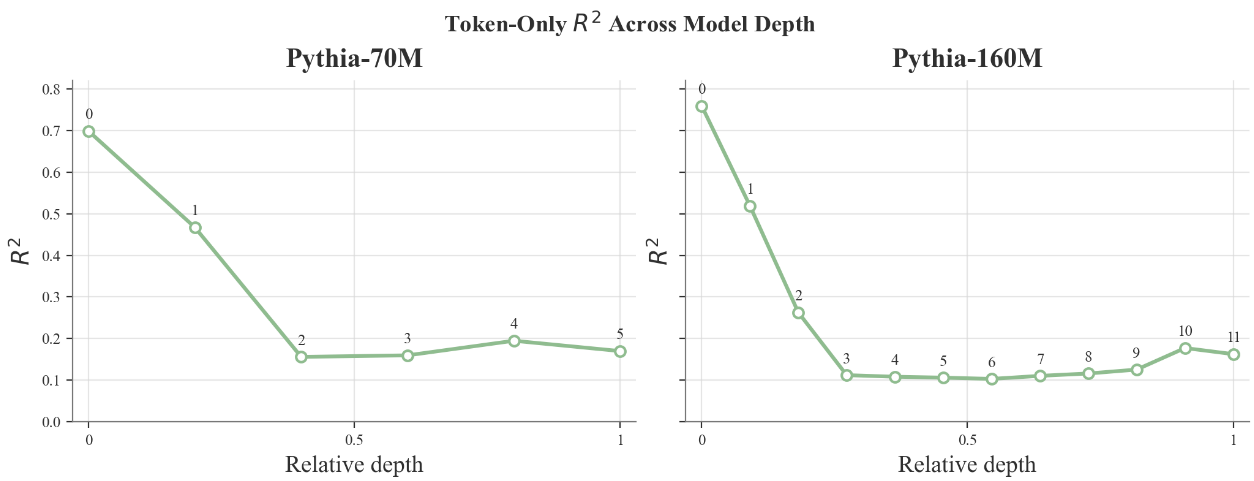

Figure 5: Side-by-side comparison of token-only probe R^2 across relative depth for Pythia-70M and Pythia-160M. Both models show high recoverability in the earliest layers followed by a sharp drop, indicating that token identity alone explains much less of the middle-layer hidden-state variance.

Both models show the same broad structure. The earliest hidden states are strongly predicted by token identity, and predictability falls sharply after the first few transformer blocks. The 160M model makes this transition easier to see: the drop is gradual from layers 0 to 2 and then nearly flat at low R^2 across the middle of the model. The 70M run has fewer layers, but it still shows the same decline by layer 2.

The most important caution is that the final-layer cosine spike appears in both models. In the 70M model, the final layer has the largest MSE but the highest cosine. In the 160M model, layer 11 behaves similarly, with MSE 2.58, R^2 0.16, mean cosine 0.99, and mean norm about 270. This makes the final-layer result a metric interaction rather than a simple success case. Direction is preserved well, but magnitude and coordinate-level reconstruction remain poor.

5 Discussion

The main result is that token identity alone explains a large fraction of early hidden-state structure but much less of the middle-layer structure. This matches the intuition that the first transformer blocks still carry strong lexical information, while later blocks have mixed in more context from previous tokens. The results do not prove that all unexplained variance is context information: some of it could be nonlinear token effects, probe misspecification, model normalization behavior, or noise from the finite sample. Still, because the probe receives no previous-token information, the sharp drop in held-out R^2 is strong evidence that the representation is no longer mostly a linear function of the current token embedding.

Several parts of the project worked well. The extraction, probing, plotting, and diagnostic pipeline is reproducible from YAML configs, and the Pythia-160M run extends the analysis beyond a toy smoke test. The main layerwise metric plots also use 10 train/test splits, which makes those trends more trustworthy than a single split alone. The diagnostics also caught an important interpretability issue: cosine similarity alone would make the final layer look almost perfectly reconstructed, while MSE and R^2 show that this would be misleading.

The main limitation is scale. These experiments use small models and relatively small token samples because the full activation tensors are expensive to store and process locally. The probe is also intentionally simple. An affine map is a clean baseline, but a poor affine reconstruction does not rule out nonlinear token-only predictability. Finally, the present report analyzes whole hidden states rather than interpretable features, so the results say where token-only predictability changes but not exactly which semantic or syntactic features become context-dependent.

6 Conclusion

This project tested how much of a transformer's hidden state can be reconstructed from the current token embedding alone. Across Pythia-70M and Pythia-160M, the answer is layer-dependent: early layers are substantially token-predictable, but middle layers are not. The Pythia-160M curve in particular suggests that most of the transition happens by about layer 3. The final layer shows strong directional alignment with poor absolute reconstruction, demonstrating why MSE, R^2, and cosine should be interpreted together.

6.1 Future Work

With more time, I would run the same experiment on larger Pythia checkpoints, add ridge-regularized and nonlinear-probe ablations, compare shuffled-context or same-token baselines, and extend the pipeline to sparse autoencoder features. The sparse-feature version would be the most interpretable next step because it would turn the current whole-vector question into a feature-level question: which features are token-local, and which require context?

References

[1] Guillaume Alain and Yoshua Bengio. Understanding intermediate layers using linear classifier probes, 2016. URL `https://arxiv.org/abs/1610.01644`.

[2] Nora Belrose, Igor Ostrovsky, Lev McKinney, Zach Furman, Logan Smith, Danny Halawi, Stella Biderman, and Jacob Steinhardt. Eliciting latent predictions from transformers with the tuned lens, 2023. URL `https://arxiv.org/abs/2303.08112`.

[3] Stella Biderman, Hailey Schoelkopf, Quentin Gregory Anthony, Herbie Bradley, Kyle O'Brien, Eric Hallahan, Mohammad Aflah Khan, Shivanshu Purohit, USVSN Sai Prashanth, Edward Raff, Aviya Skowron, Lintang Sutawika, and Oskar Van Der Wal. Pythia: A suite for analyzing large language models across training and scaling. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 2397–2430. PMLR, 2023. URL `https://proceedings.mlr.press/v202/biderman23a.html`.

[4] Trenton Bricken, Adly Templeton, Joshua Batson, Brian Chen, Adam Jermyn, Tom Conerly, Nicholas L. Turner, Cem Anil, Carson Denison, Amanda Askell, Robert Lasenby, Yifan Wu, Shauna Kravec, Nicholas Schiefer, Tim Maxwell, Nicholas Joseph, Zac Hatfield-Dodds, Alex Tamkin, Karina Nguyen, Brayden McLean, Josiah E. Burke, Tristan Hume, Shan Carter, Tom Henighan, and Christopher Olah. Towards monosemanticity: Decomposing language models with dictionary learning. *Transformer Circuits Thread*, 2023. URL `https://transformer-circuits.pub/2023/monosemantic-features/`.

[5] Nelson Elhage, Neel Nanda, Catherine Olsson, Tom Henighan, Nicholas Joseph, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, Tom Conerly, Nova DasSarma, Dawn Drain, Deep Ganguli, Zac Hatfield-Dodds, Danny Hernandez, Andy Jones, Jackson Kernion, Liane Lovitt, Kamal Ndousse, Dario Amodei, Tom Brown, Jack Clark, Jared Kaplan, Sam McCandlish, and Chris Olah. A mathematical framework for transformer circuits. *Transformer Circuits Thread*, 2021. URL `https://transformer-circuits.pub/2021/framework/`.

[6] Leo Gao, Stella Biderman, Sid Black, Laurence Golding, Travis Hoppe, Charles Foster, Jason Phang, Horace He, Anish Thite, Noa Nabeshima, Shawn Presser, and Connor Leahy. The pile: An 800gb dataset of diverse text for language modeling, 2020. URL `https://arxiv.org/abs/2101.00027`.

[7] John Hewitt and Christopher D. Manning. A structural probe for finding syntax in word representations. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 4129–4138, Minneapolis, Minnesota, 2019. Association for Computational Linguistics. doi: 10.18653/v1/N19-1419. URL `https://aclanthology.org/N19-1419/`.

[8] Catherine Olsson, Nelson Elhage, Neel Nanda, Nicholas Joseph, Nova DasSarma, Tom Henighan, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, Tom Conerly, Dawn Drain, Deep Ganguli, Zac Hatfield-Dodds, Danny Hernandez, Andy Jones, Jackson Kernion, Liane Lovitt, Kamal Ndousse, Dario Amodei, Tom Brown, Jack Clark, Jared Kaplan, Sam McCandlish, and Chris Olah. In-context learning and induction heads, 2022. URL `https://arxiv.org/abs/2209.11895`.

[9] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems 30*, 2017. URL `https://proceedings.neurips.cc/paper/2017/hash/3f5ee243547dee91fbd053c1c4a845aa-Abstract.html`.

A Reproducibility Details

The main experiments use the following configuration files:

- Pythia-70M: configs/pythia70m.yaml
- Pythia-160M: configs/pythia160m.yaml

The seed-0 split is used for the tables and diagnostic examples in the report. The main layerwise metric plots in Figures 1 and 3 use 10 train/test splits with seeds 0 through 9, and their shaded regions show one standard deviation across splits. No ridge regularization is used. The generated metrics are stored in results/generated/, and the figures used in this report are stored under the final report figures directory.

B Additional Diagnostic Figures

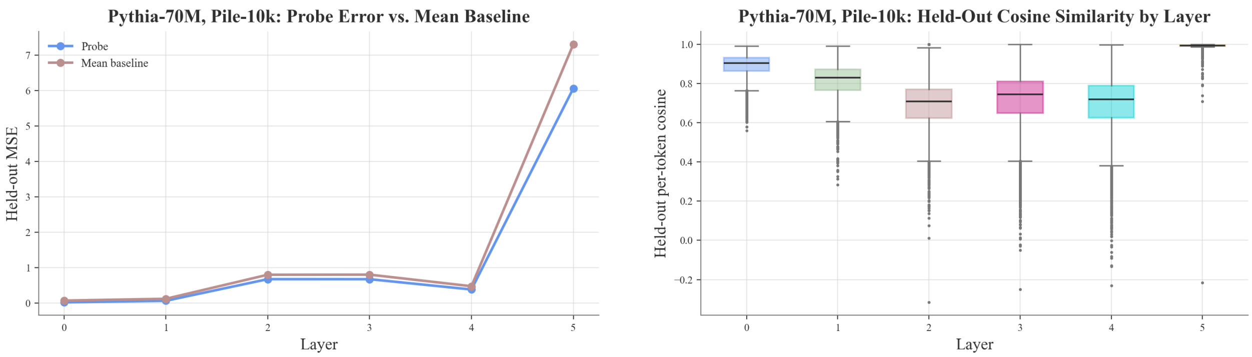

Figure 6: Additional Pythia-70M diagnostics. Left: held-out probe MSE compared with a constant mean-hidden-state baseline. Right: distribution of per-token cosine similarities by layer.

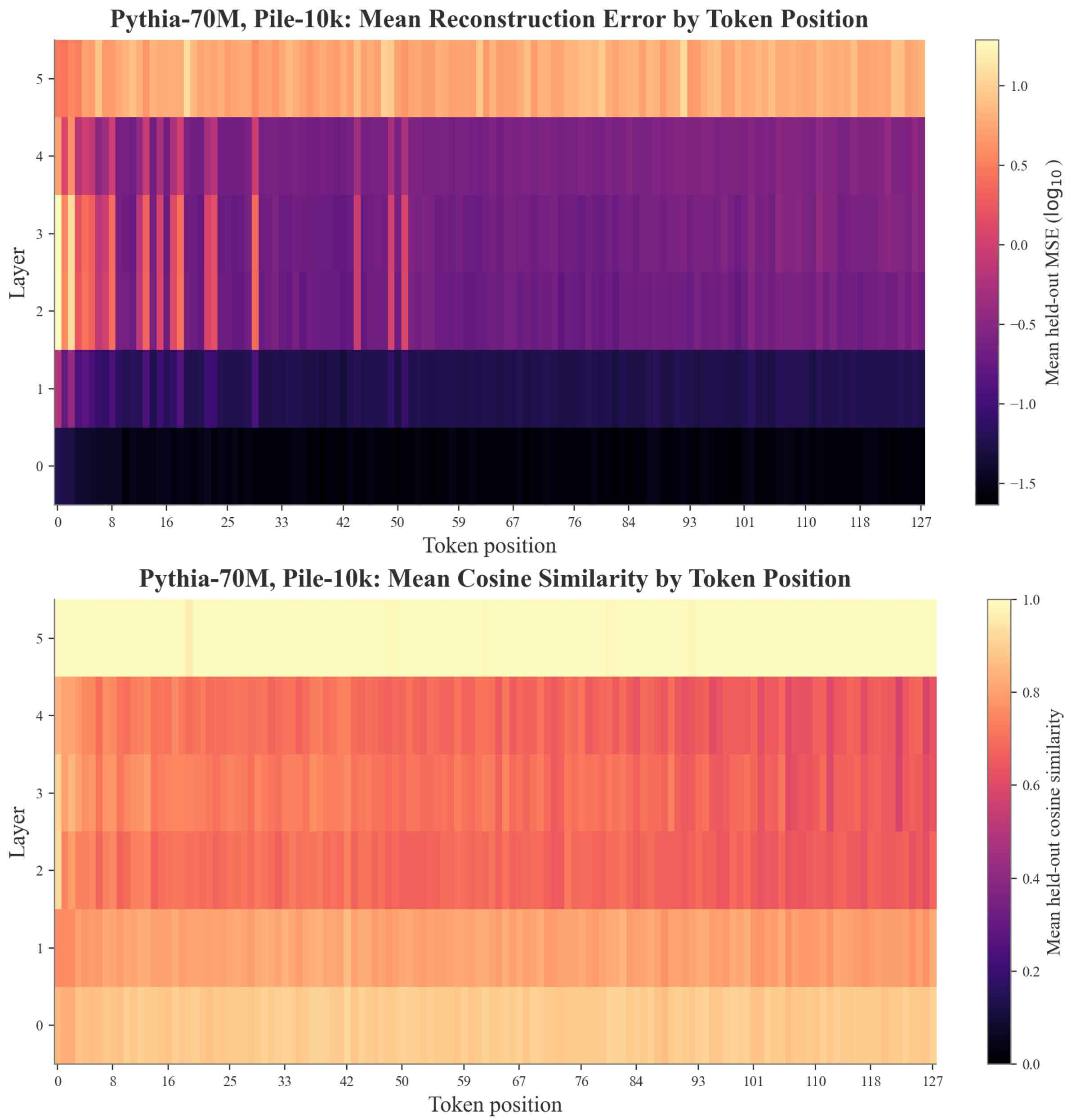

Figure 7: Pythia-70M token-level diagnostics. Top: reconstruction error by sequence position. Bottom: token-by-layer error heatmap for sampled held-out tokens.

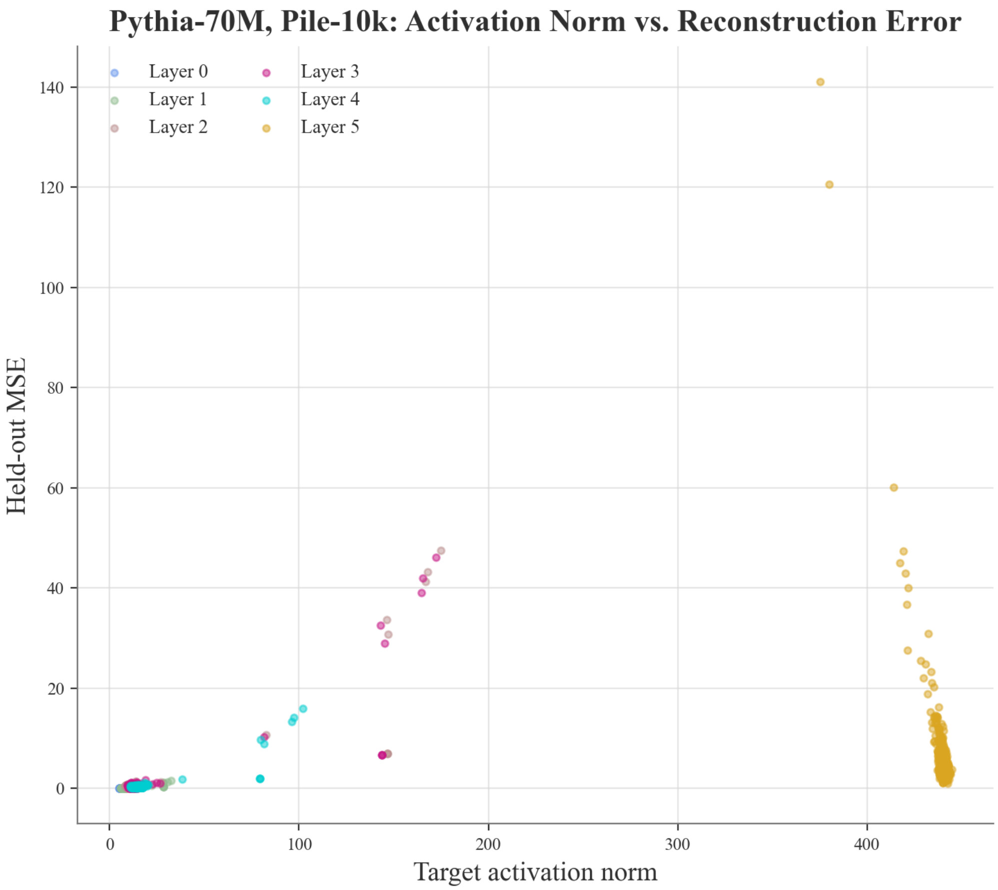

Figure 8: Pythia-70M relationship between hidden-state norm and token-only reconstruction error.

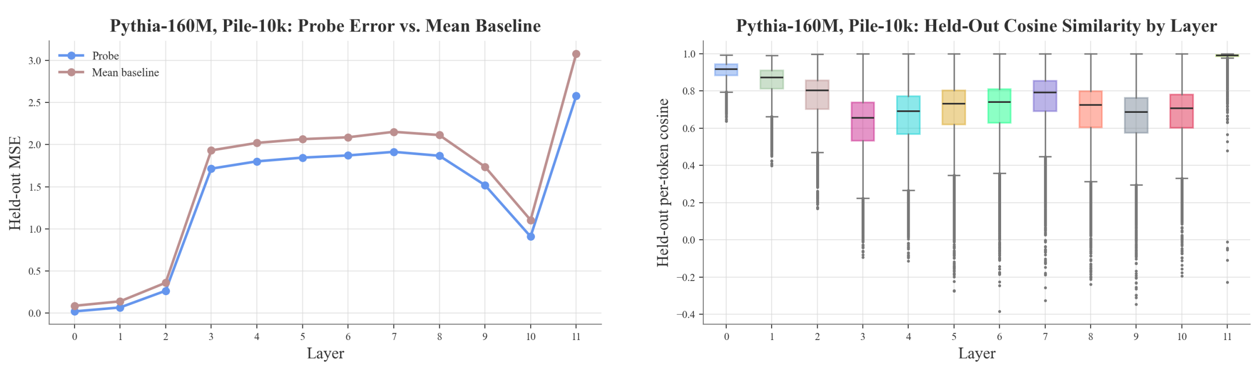

Figure 9: Additional Pythia-160M diagnostics. Left: held-out probe MSE compared with a constant mean-hidden-state baseline. Right: distribution of per-token cosine similarities by layer.

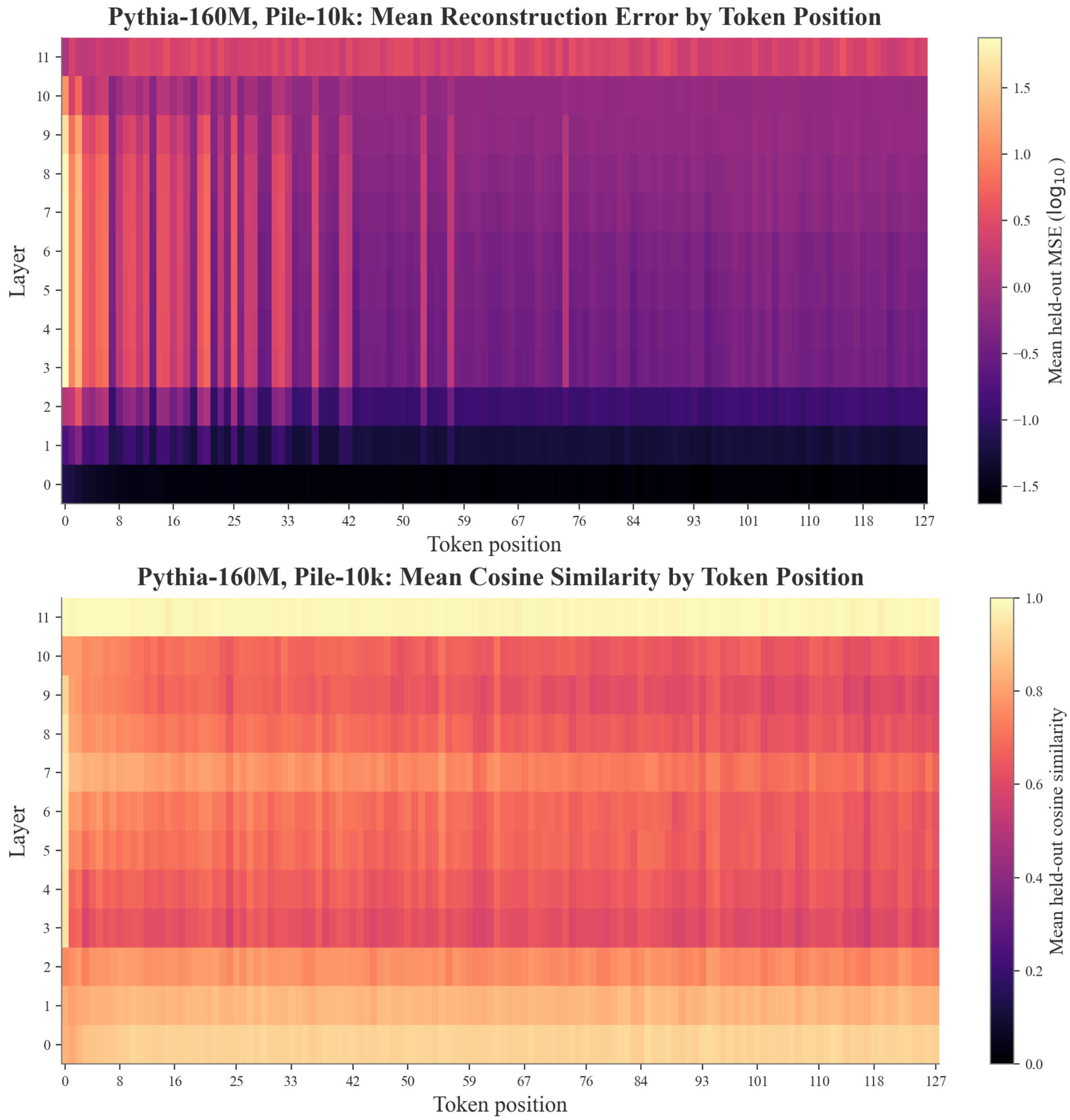

Figure 10: Pythia-160M token-level diagnostics. Top: reconstruction error by sequence position. Bottom: token-by-layer error heatmap for sampled held-out tokens.

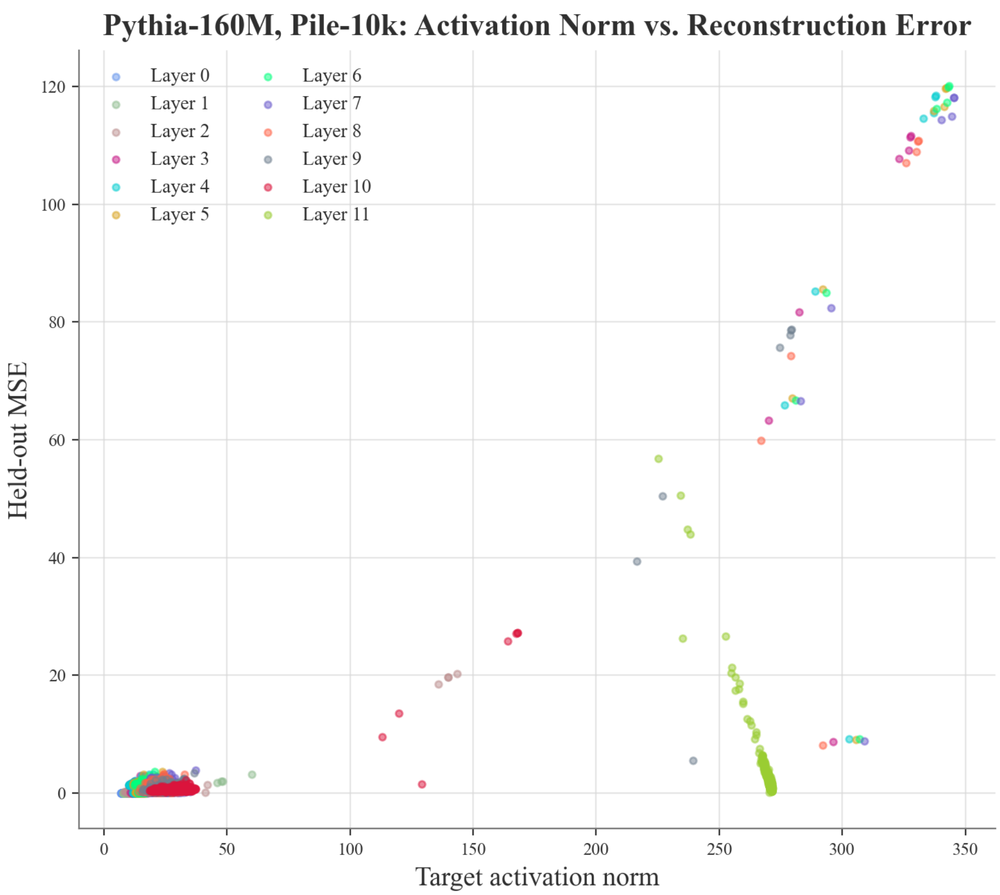

Figure 11: Pythia-160M relationship between hidden-state norm and token-only reconstruction error. The final-layer norm increase explains why cosine similarity can be high even when MSE remains large.